Chemosphere 367 (2024) 143639

Contents lists available at ScienceDirect

Chemosphere

journal homepage: www.elsevier.com/locate/chemosphere

Role of polylactic acid microplastics during anaerobic co-digestion of cow manure and Chinese cabbage waste enhanced by nanobubble

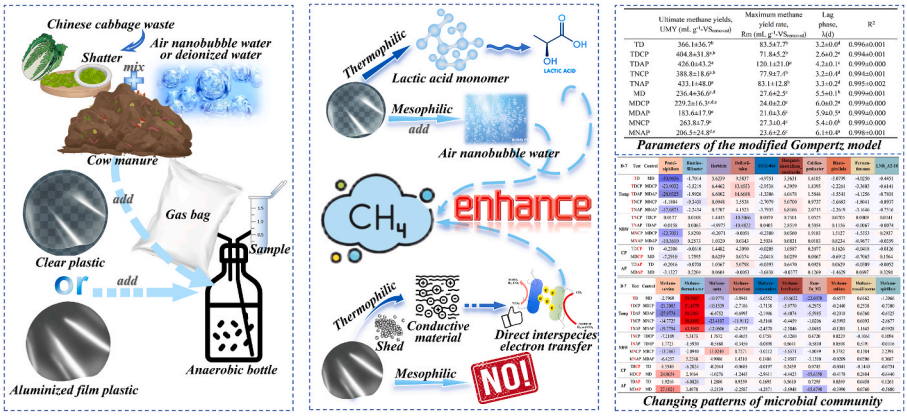
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HIGHLIGHTS

- PLA promoted AD in thermophilic condition with a 6.2–18.3% increase in methane yield.
- PLA was resistant to mesophilic AD and decreased methane yield.
- Aluminised film acted as a conductive material in thermophilic AD.
- Added air-NBW to PLA-added digesters can increase methane yield by 12.5–15.1%.
- *Methanothermobacter* was highest in the whole AD process when PLA and NBW were added.

GRAPHICAL ABSTRACT



ARTICLE INFO

Handling editor: Grzegorz Lisak

Keywords:
Clear microplastic
Aluminised film microplastic
Hydrolysis of microplastics
Lactic acid
Inhibition
Electrically conductive material

ABSTRACT

With the increasing use of plastic products globally, environmental pollution by plastic waste is becoming increasingly problematic. This study investigated the impacts of two types of polylactic acid microplastics, clear microplastics and aluminised film microplastics, on methane yield, microbial community, and volatile fatty acid accumulation during anaerobic co-digestion of cow manure and Chinese cabbage waste under different temperature conditions. The influence of the addition of air nanobubbles on microplastic degradation in the anaerobic digestion system we also examined. The results revealed that under thermophilic conditions, clear and aluminised film microplastics increased the methane yield, with the latter resulting in greater improvement. Conversely, under mesophilic conditions, the presence of microplastics reduced the methane yield, but the addition of air-nanobubble partially mitigated this effect. Microplastics also affected the microbial community, with specific species showing correlations with methane yield. *Methanothermobacter*, which is linked to lactic acid conversion, was positively correlated with methane yield, whereas *Methanomassiliicoccus* levels increased in the presence of microplastics, particularly in the inhibited state of the digester. These results suggest that, under thermophilic conditions, microplastics may increase the cumulative methane yield by facilitating the

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https://doi.org/10.1016/j.chemosphere.2024.143639
Received 7 July 2024; Received in revised form 16 October 2024; Accepted 24 October 2024
Available online 4 November 2024
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degradation of lactic acid monomers. Furthermore, the aluminised film on microplastics could serve as an electrically conductive material during anaerobic digestion, potentially increasing the methane yield.

1. Introduction

Co-digestion of cow manure (CM) and Chinese cabbage waste (CCW) can increase anaerobic digestion (AD) system stability and methane yield by encouraging microbial diversity and synergistic effects (Liu et al., 2022). AD efficiency is strongly influenced by the makeup of CM and CCW, which may contain a high proportion of difficult-to-degrade substances or form toxic compounds (Zhang et al., 2024).

In addition, plastics are increasingly used globally and the fact that many agricultural services are provided by plastics (Liu et al., 2014), some of which degrade into microplastics (MPs) over time because of physical, chemical and biological reactions. These MPs subsequently find their way into livestock waste through processes such as animal feeding, manure composting, and other means (Wu et al., 2021a,b; Zhang et al., 2020). MPs are plastic fragments and particles with a diameter of less than 5 mm The biofilms present on MPs can serve as carriers for pathogens and may also adsorb heavy metals (Lu et al., 2020). Fecal matter containing MPs can introduce these harmful substances into farmlands, which in turn have the potential to enter the human body via crops or other pathways, thereby posing a threat to human health (Lu et al., 2020). MPs were detected in CM at an abundance of 74 items per kilogram of wet matter (Sheriff et al., 2023).

MPs may adversely affect AD under short-term and long-term exposure, and different MPs may affect AD depending on their physical and chemical properties (Mohammad Mirsoleimani Azizi et al., 2021). Petroleum-based plastics such as polyvinyl chloride MPs (50 nm) inhibit methane yield by hindering acid production during AD. They also induce oxidative stress, which reduces cell viability (Wang et al., 2022a). Polyethylene MPs can inhibit diversity and affect interspecies electron transfer and related genes in methanogens, attenuating related enzymes and electron transfer activities, and further reducing methane yield rates (Wang et al., 2022b). Howere, biobased plastics such as polylactic acid plastic (PLA), which is derived from the fermentation of starch or sugarcane and has excellent degradation properties, utilised thermophilic methanogens under thermophilic conditions to degrade lactic acid polypeptides while completing the methane conversion, which can promote methane yield (Bandini, 2020; Shang et al., 2023). However, PLA degradation under mesophilic conditions is different from that under thermophilic conditions, and the PLA copolymer can be completely degraded in 116 ± 48 days at 55 °C and in 423 ± 76 days at 37 °C (Cazaudehore et al., 2021; Lu et al., 2022; Zhang et al., 2018).

Biodegradable plastics are used as alternatives to traditional fossil-source plastics because of their biodegradability under specific conditions (Peng et al., 2024). PLA is the world's highest-capacity bioplastic, accounting for 31.0% of bioplastic capacity and is expected to account for 43.6% by 2028 (EUBP, 2023). PLA is used in a wide range of applications such as product packaging, construction, and agriculture (Jamshidian et al., 2010). PLA in practical applications, to cater to the characteristics required by the market, is usually further processed (Jamshidian et al., 2010). In agricultural production, aluminised mulch is widely used for pest control (Croxton and Stansly, 2014), so it often appears in agricultural waste and ends up in livestock manure (Li et al., 2014). Aluminised films use the excellent physicochemical properties of aluminium to enhance the barrier properties, light reflection properties and antistatic capabilities of plastic films, thereby improving their functionality (Mwema et al., 2018). With the gradual and widespread use of PLA, the prospect of aluminised films on PLA surfaces, such as aluminised PLA mulch, which has a high probability of appearing in agricultural waste, livestock manure, etc., will be very wide.

Therefore, simple, economical, and efficient methods are needed to reduce the negative impact of PLA-MPs on AD while improving hydrolysis and stabilising the digester during AD. Nanobubbles (NBs) can improve substrate accessibility, digestibility and enzyme activity due to their small size, high electrostatic interactions and ability to generate reactive oxygen species (Chuenchart et al., 2021). Therefore, the addition of nanobubble water (NBW) to AD to facilitate the resourceful utilisation of substrates has gradually gained attention. The addition of NBW increased the hydrolysis and acidification processes of AD, and the reactive oxygen species generated by NBW promoted the hydrolysis of lignin by breaking its encapsulated structure, which increased the degradation rate of lignin by 19.0%, and simultaneously increased the abundance of hydrolysing bacteria and methanogens by 52.9% and 63.8%, respectively (Yang et al., 2020). NBW can increase water mobility, allowing more nutrients to be transferred to microbial cells and increasing the production of extracellular enzymes, which increase the activity of extracellular hydrolytic enzymes by 14.0%–17.0% and those related to lignocellulose decomposition by 38.0% (Wang et al., 2020). NBW creates a trace amount of oxygen to increase the efficiency of the electron transport system by a factor of 1.7, resulting in a VFA (Volatile Fatty Acid) reduction of 193 NmL g^{-1}VS by enhancing the activity of parthenogenetic bacteria (Wang et al., 2021a). Among the multiple types of NBW (air, CO_2, He and N_2), air-NBW is the most cost-effective, increasing proteolysis and resulting in a 27.8% increase in the cumulative methane yield (Wang et al., 2024). In addition, the combination of MPs and NBs resulted in an increase in the measured mean particle size and concentration, while the NBs decreased the aggregation between MPs, and the presence of NBs bound to MPs made them more stable in the anaerobic environment (Wang et al., 2023). However, studies of NBW and MPs acting together in AD are more limited, and questions about whether NBW or MPs have a greater effect on AD, and whether the simultaneous presence of NBW and MPs affects AD differently are currently unknown. Additionally, how biobased MPs promote or hinder AD is insufficiently studied, and how to amplify their promoting effect or alleviate their hindering effect is unclear.

Therefore, this study focuses on clear PLA (CP) and aluminised film PLA (AP) as representatives to explore how the presence of biobased MPs affects AD during anaerobic co-digestion of CM and CCW, and whether the facilitation of AD by NBW is affected by the presence of PLA-MPs. To assess the stability of AD, the pH, soluble chemical oxygen demand (SCOD), volatile fatty acid (VFA), total ammonia nitrogen (TAN), and CMY of the AD system were determined in this study.

2. Materials and methods

2.1. Preparation of NBW

The air-NBW used in this study was prepared via a nanobubble generator (JXWNP-AL01, Jinxiang, China). Air was introduced into deionized water (DW) through this generator for 20 min at 20 ± 2 °C and left for 10 min to change the liquid from milky to clear to obtain the NBW for the whole AD process. The pressure of the nanobubble generator was controlled between 0.4 and 0.5 MPa during the preparation of NBW. After preparation, NBW was used in the subsequent AD.

2.2. Materials

CM and CCW were collected from a dairy farm and wholesale vegetable market in Lanzhou, China. There are two types of PLA plastic: CP and AP. CP is a film with a thickness of 25 μm and a smooth surface, and the AP is a film with a thickness of 0.1 nm that is plated on CP via a vacuum single-sided aluminising method. According to the measurement, the conductivity of the aluminised film is 2991.5 S m^{-1}, which is

electrically conductive (ST2253y, Suzhou, China). PLA is purchased from Huixing Environmental Protective Material Co., Ltd. (China). In the laboratory, circles with a diameter of 1.5 ± 0.1 mm were cut via a 1.5 mm hole punch (4120, Tianhong, China) (Lu et al., 2022). CM and CCW were crushed with a pulveriser and stored at 4 ± 1 °C in advance. The inoculums used in the experiment were domesticated thermophilic and mesophilic digested sludge. The Supplementary material section lists the major characteristics of the different substrates used in this study.

2.3. Experimental design

The experimental setup and sampling methods were similar to those used in previous experiments (Liu et al., 2022; Zhang et al., 2024). A digester with a working volume of 1 L was used; on the basis of previous experiments, the optimal substrate mixing ratio was selected. The optimal substrate ratio with the best hydrolysis effect and higher methane yield efficiency, CM:CCW = 1:1 (wet basis) (Liu et al., 2022), for the best hydrolysis and higher methane yield efficiency in general was selected, whereas the substrate (CM + CCW) and water ratio = 1:1 (Zhang et al., 2024) were chosen on the basis of the optimal total solids (TS) for AD, resulting in a TS of approximately 10% for the reactor (Abid et al., 2021). A total of 700 g of CM, CCW and water was finally loaded. Equal amounts of CP and AP were added to the anaerobic digester mixed with NBW or DW at a concentration of 5.0 ± 0.1 g Kg^{-1}TS, respectively. The concentration of MPs was referenced from Lu et al. (2022), which was a minimum concentration of MPs close to the range of the global level (Itävaara et al., 2002). This experiment uses two temperatures, thermophilic (T, 55 ± 1 °C) and mesophilic (M, 37 ± 1 °C), with a total of ten conditions, which are TD (T + DW), TDCP (T + DW + CP), TDAP (T + DW + AP), TNCP (T + NBW + CP), TNAP (T + NBW + AP), MD (M + DW), MDCP (M + DW + CP), MDAP (M + DW + AP), MNCP (M + NBW + CP), and MNAP (M + NBW + AP). Details are provided in Table 1, where TD and MD served as the control group. Subsequently, 70 g of thermophilic or mesophilic seed sludge was added to the digesters. Three sets of parallel experiments were set up for each set. A portion of the digestion mixture was pumped out by a peristaltic pump at the same weekly intervals and centrifuged at 2000 *g* for 5 min. The supernatant was removed and filtered through a 0.45 μm polyether sulfone microfibre filter, and the filtered liquid was used for experimental testing. The pH was not artificially adjusted in this study.

2.4. Analytical methods

TS, volatile solids (VS), pH, TAN, SCOD, and total alkalinity (TA) were determined via the standard method (APHA, 1998). The free ammonia nitrogen (FAN) content was calculated via the equation provided by Hansen et al. (1998). The VFA was determined via a gas chromatograph (SP-7890 Plus, Ruihong, China) with an organic acid analytical column (320 μm inner diameter, 0.50 μm film, and 30 m total length) and a flame ionization detector. The volume of biogas was measured via a gas flowmeter (LML-2, Alpha, China), and the methane concentration in the biogas was measured via a biogas composition analyser (GA5000, Geotach, United Kingdom).

Deoxyribonucleic acid (DNA) extraction was performed at fixed times, and the microbial community was analysed by sequencing the V3–V4 and V4–V5 hypervariable regions of the 16S rDNA gene. Bacteria were amplified with primers 341F (CCTACGGGGNGGCWGCAG) and 806R (GGACTACHVGGGGTATCTAAT), and archaea were amplified with primers Arch519F (CAGCMGCCGCGGGTAA) and Arch915R (GTGCTCCCCCGCCAATTCCT). Sequencing was then performed on the Illumina MiSeq platform (AxyPrep DNA Gel Extraction Kit, Axygen Biosciences, Union City, CA, U.S.), and microbial abundance and typology analyses were conducted to determine differences in the microbial community in each digester.

Table 1
Experimental group operating condition setting.

Factors	Test	Control
Thermophilic(T)/Mesophilic(M)	**T**D	**M**D
	TDCP	**M**DCP
	TDAP	**M**DAP
	TNCP	**M**NCP
	TNAP	**M**NAP
Deionized water (D)/Nanobubble water (N)	T**N**CP	T**D**CP
	T**N**AP	T**D**AP
	M**N**CP	M**D**CP
	M**N**AP	M**D**AP
Clear plastic (CP) or not	TD**CP**	TD
	MD**CP**	MD
Aluminised film plastic (AP) or not	TD**AP**	TD
	MD**AP**	MD
Clear plastic (CP)/Aluminised film plastic (AP)	TD**AP**	TD**CP**
	TN**AP**	TN**CP**
	MD**AP**	MD**CP**
	MN**AP**	MN**CP**

*The bolded font in the table represents the control for each factor.

2.5. Statistical analysis

In this study, all the statistical analyses were performed via SPSS 27.0 version software for Windows. One-way ANOVA was used to evaluate the effects of different experimental conditions on the outcome variables. Duncan's Multiple Range Test, T-test and Dunnett's T3 test were used to determine the differences between the different samples with a confidence level of 95%. The data are shown as the means ± standard deviations. $P < 0.05$ was considered statistically significant (Lu et al., 2022). CMY was fitted using a modified Gompertz model (Liu et al., 2022).

3. Results and discussion

3.1. Digestion performance

3.1.1. Characteristics of the digestate

The pH (Fig. 1a and b) is an essential indicator of AD, and a suitable pH for methanogens is in the range of 6.5–8.0 (Li et al., 2017). During AD, the pH under thermophilic conditions was greater than that under mesophilic conditions, probably because temperature affects microbial processes by influencing the nature of metabolism, nutrient requirements, biomass composition, and reaction rates (Maleki et al., 2018). Thermophilic conditions promote hydrolysis, which breaks down VFA in time and does not result in the accumulation of VFA (Li et al., 2023; Zhang et al., 2024). The pH slightly decreased in the initial stage, probably because PLA was hydrolysed as a substrate to produce VFA when it was added to the AD digester, which led to an increase in VFA and a decrease in pH (Seeley et al., 2020). However, the pH under mesophilic conditions decreased more than that under thermophilic conditions in the initial stage, probably because acid-producing bacteria adapt to the environment faster than methanogens do and grow faster resulting in the accumulation of VFA from PLA (Clagnan et al., 2023). In addition to VFA, pH is also affected by the amount of ammonia present in the digester (Wang et al., 2024).

Ammonia is usually produced by the hydrolysis of proteinaceous substances in the substrate (Liu et al., 2022). A reasonable concentration of ammonia can act as a buffer and provide nutrients for microorganisms (Zhang et al., 2024). Under thermophilic and mesophilic conditions, TAN (Fig. 1c and d) and FAN (see **supplementary material**) gradually increased during the digestion process. The activity of methanogens was significantly inhibited when the TAN was greater than 1500 mg L^{-1} or the FAN was greater than 250 mg L^{-1} (Liu et al., 2022). FAN is more toxic to methanogens than TAN, and FAN can penetrate microbial cell membranes and destroy enzyme systems (Liu et al., 2022). The TAN and FAN in the thermophilic digester were significantly greater than those in

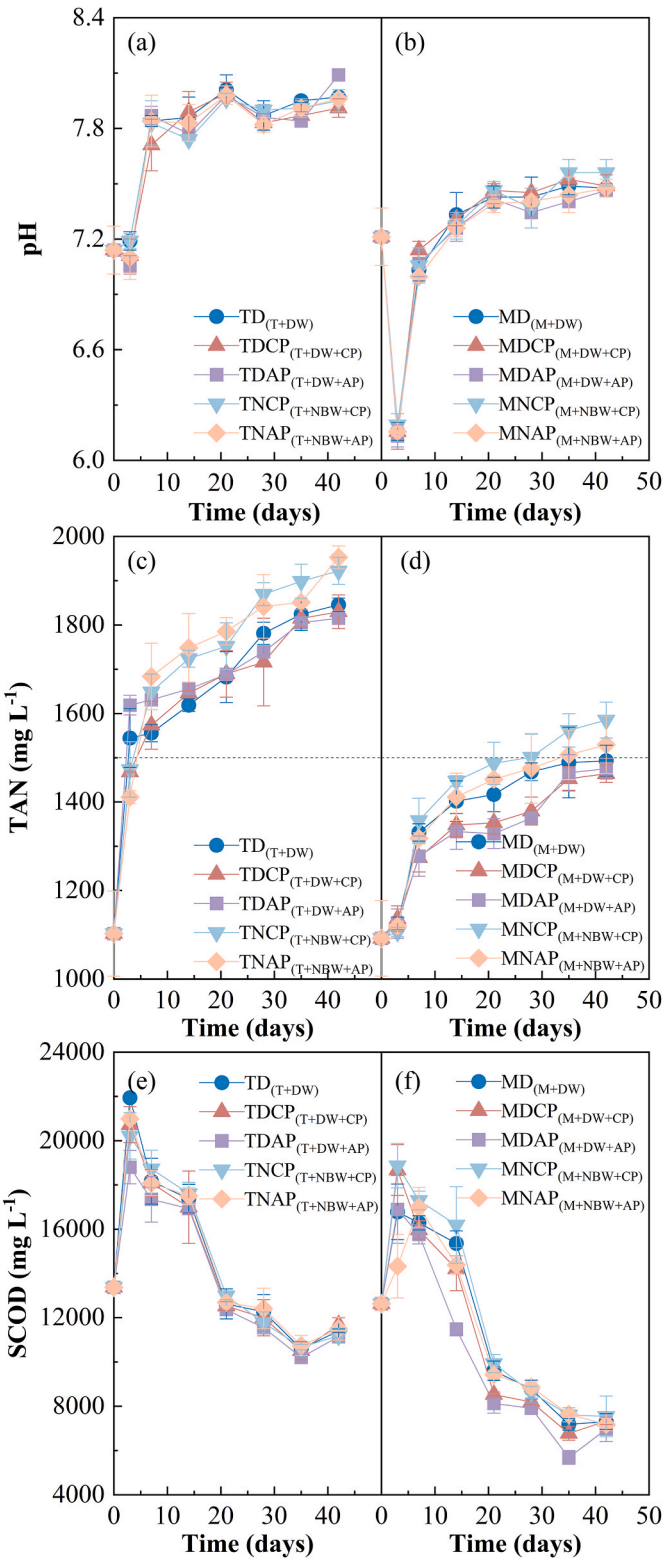


Fig. 1. Changes in pH (a and b), TAN (c and d), SCOD (e and f) during the digestion process.

the mesophilic digester ($P < 0.01$).

At the end of the digestion process, the TAN and FAN in the NBW-unadded but PLA-added digesters were lower than those in the control group, suggesting that the addition of PLA-MPs may have affected the hydrolysis of AD to be negatively and that the addition of MPs caused the gene expression of hydrolysis-associated microorganisms to be impeded (Wang et al., 2022b). In contrast, the digesters with the addition of NBW presented higher TAN than the control group, and the TNCP and TNAP increased by 4.1% and 5.8%, respectively. The TAN of TNCP and TNAP were significantly greater than that of TD on Day 7 ($P < 0.01$), with increases of 6.0% and 8.2%, respectively. These findings indicate that, in addition to temperature, NBW also plays a role in protein hydrolysis (Zhang et al., 2024). The promotion of AD hydrolysis by NBW was greater than the hindering effect of MPs, and the presence of MPs did not affect the NBW promotion of the hydrolysis stage. TAN shows an increasing trend, in contrast to SCOD, which usually shows an overall increasing and then decreasing trend (Wang et al., 2024).

SCOD (Fig. 1e and f) is an essential indicator of the amount of dissolved reducing substances in the digestate (Liu et al., 2022). On Day 3, SCOD peaked under thermophilic and mesophilic conditions. Nevertheless, the SCOD level under thermophilic conditions was significantly greater than that under mesophilic conditions ($P < 0.01$), and the TD was 23.5% greater than that under MD, which might be due to the complete hydrolysis and release of more VFA under thermophilic conditions.

Under thermophilic conditions, SCOD levels peaked on Day 3. The SCOD level of TD was the greatest and the TDAP level was the lowest, the TDAP level was 14.3% lower than that of TD, and there was a significant difference ($P < 0.01$). It is speculated that the AD reactor was already in the methanogenic stage at this time, and SCOD had begun to decline before sampling. This finding also suggest that the addition of AP increased the suitability of methanogens for growth in the digester (Liang et al., 2023), therefore, the SCOD level decreased the fastest. Under mesophilic conditions, at the peak on Day 3, the SCOD levels were MNCP, MDCP, MDAP, MD, and MNAP in descending order. The SCOD level of the CP-added digesters were significantly greater than that of the AP-added digesters and significantly greater than that of the control digesters ($P < 0.01$), suggesting that the CP promoted hydrolysis, perhaps because CP was used as a substrate hydrolysed to increase the VFA in the system, whereas the aluminised film on the surface of the PLA physically impeded the hydrolysis of the PLA (Cazaudehore et al., 2022), resulting in the failure of the VFA release.

In general, VFA (Fig. 2), as a major component of SCOD, is a product of hydrolysis and acidification and is an essential intermediate during the AD (Li et al., 2017). On Day 3, the VFA levels of the TDCP and TNCP digesters were 12.9% and 21.2% greater than those of the control group, respectively. It can be assumed that CP was hydrolysed as additional carbonaceous organic matter in AD to produce more VFA (Seeley et al., 2020). In contrast, the VFA levels of TDAP and TNAP were 15.0% and 10.2% lower than those of TDCP and TNCP, respectively, suggesting that the methanogens grew more appropriately in the AP-added digester. During AD, the aluminised film of PLA was shed as PLA underwent hydrolysis because of its inherent conductivity, and the fallen aluminium functioned as the conductive material. Conductive materials promote the growth of bacteria by increasing the specific surface area and promoting the conversion of VFA by methanogens by enhancing the interspecies electron transfer (Liang et al., 2023). Furthermore, the VFA levels in the NBW-added digester were 12.4%–18.7% greater than those in the DW-added digester, which indicates that the addition of NBW leads to more efficient hydrolysis in the reaction, more complete substrate degradation, and faster conversion of VFAs to methane (Zhang et al., 2024). This phenomenon is consistent with the results of subsequent studies (Fig. 3 and Table 2). It is speculated that the trace oxygen environment created by air-NBW promotes the expression of hydrolysis-associated parthenogenetic bacterial genes, thus allowing for an increase in VFA production from hydrolysis (Chuenchart et al., 2021).

Under mesophilic conditions, the VFA levels of the MPs-added digester were greater than those of the control group, suggesting that the unhydrolysed MPs provided a greater specific surface area for the microbes to survive at a physical level, which allowed the more adaptable acid-producing bacteria to multiply faster, resulting in the

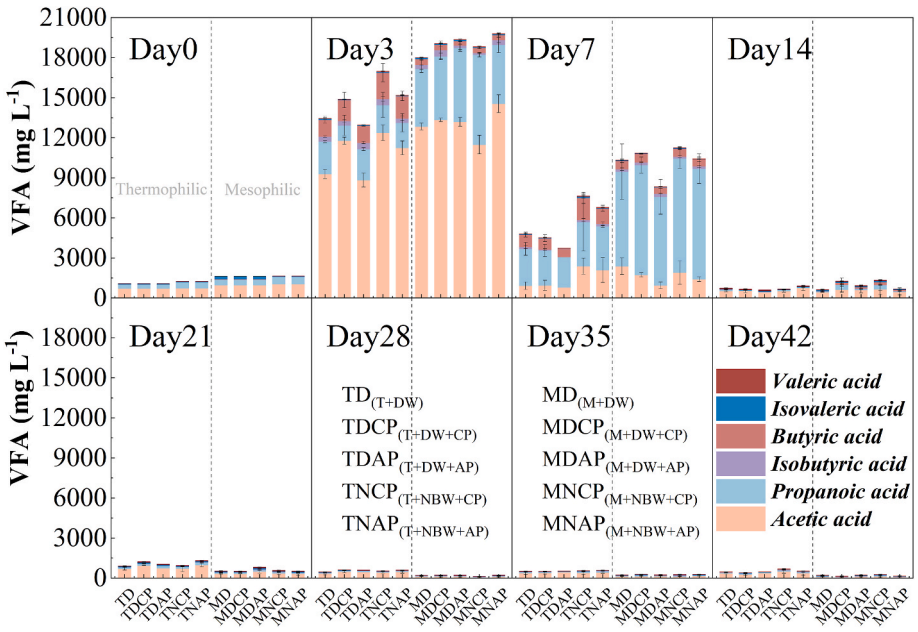


Fig. 2. Changes in VFA during the digestion process.

accumulation of VFA (Clagnan et al., 2023; Lee et al., 2022). The VFA level of MNAP was 3.2% greater than that of MDAP, so it was speculated that the hindering effect of AP on AD was less than the promoting effect of NBW, and that while AP provided the specific surface area for the microorganisms, NBW promoted hydrolysis to produce VFA either by increasing the mobility of water or by increasing the activity of hydrolytic enzymes (Chuenchart et al., 2021; Wang et al., 2024). The VFA levels were lower on Day 7 than on Day 3. The VFA levels in the AP-added digester decreased even further, indicating that the aluminised film promoted the methane yield. The addition of aluminium can promote direct interspecies electron transfer between fermentative bacteria and methanogens, playing a positive role in maintaining the stability of the AD system and increasing the rate of methane yield, accelerating the process of converting accumulated VFA by methanogens (Liang et al., 2023).

In addition, the VFA levels in the digesters with PLA and NBW were greater than those in the digester with only PLA, suggesting that NBW can promote hydrolysis and acidification (He et al., 2023; Zhang et al., 2024). Nevertheless, the VFA in thermophilic and mesophilic digesters during AD were lower than those reported in the last survey (Liu et al., 2022). This may be due to the different natures of the AD substrates and the influence of PLA on the AD (Fan et al., 2020; Li et al., 2017; Liu et al., 2022).

3.1.2. *Methane yields*

The CMY was fitted with a modified Gompertz model (Fig. 3 and Table 2) to obtain the ultimate methane yield (UMY). The CMY (Fig. 3a and b) and daily methane yield rate (Fig. 3c and d) under thermophilic conditions were significantly higher than those under mesophilic conditions ($P < 0.01$) because thermophilic conditions promoted hydrolysis during AD, which resulted in the complete hydrolysis of the substrate, greater utilisation of organic matter than that under mesophilic conditions, and an increase in microbial activity, which offset the greater toxicity of FAN because of the increase in temperature (Liu et al., 2022). The fitting results revealed that the lag phase (λ) under thermophilic conditions was significantly smaller than that under mesophilic conditions ($P < 0.01$), suggesting that thermophilic conditions increased the solubility of organic matter, accelerated biological and chemical processes, and reduced the residence time (Fan et al., 2021).

Under thermophilic conditions, TNAP, TDAP, TDCP, and TNCP produced 18.3%, 16.4%, 10.6%, and 6.2% more methane than TD did, respectively, and TNAP and TDAP significantly increased in CMY compared with the TD ($P < 0.01$).

Compared with the control digester, the digester containing PLA-MPs produced more methane. PLA-MPs can be hydrolysed into lactic acid molecules as organic carbon substrates in AD, which leads to a yield of biomethane and improves the CMY (Clagnan et al., 2023). In addition, the CMY in the AP-added digester was greater than that in the CP-added digester, suggesting that not only the MPs but also the aluminised film on the MPs played a role in CMY in the AP-added digester. It can be assumed that the aluminised film is a type of conductive material that enhances AD through the following mechanisms: (1) it promotes direct interspecies electron transfer between substrate-oxidizing bacteria and methanogens, thus accelerating the CMY process (Yu et al., 2022); (2) it provides a high specific surface area to promote cell attachment and bacterial colonisation, thus increasing cell activity (Liang et al., 2023); (3) it enhances buffering capacity and lowers the oxidation-reduction potential (Wu et al., 2020); and (4) the aluminised film physically prevents PLA from being completely hydrolysed, so the hydrophobic surface of PLA can aid in AD by adsorbing and retaining lipids and other hydrophobic compounds, leading to better microbial metabolism and therefore more CMY (Lee et al., 2022). Therefore, the TNAP group had the maximum CMY. However, the effect of NBW on methanogens under thermophilic conditions did not show a clear pattern, and it was hypothesised that the promotion of AD methanogens by temperature was much greater than that of NBW, and that high temperature was the most important factor promoting methanogens under thermophilic conditions (Wang et al., 2022b). It is also possible that the thermophilic conditions caused the NBs to break easily, and the dissolution efficiency was reduced, which prevented the NBW from playing a role (Chuenchart et al., 2021).

Under mesophilic conditions, the CMY of MNCP was 11.6% greater than that of MD. The CMY of MDAP, MNAP, and MDCP were 22.3%, 12.6%, and 0.1% lower, respectively, than that of MD, with MNCP showing a significant increase ($P < 0.01$) in CMY compared with MD, and MDAP showed a significant decrease ($P < 0.01$) in CMY compared with MD.

The CMY of the MPs-added digester was lower than that of the control group, which may be attributed to the following reasons: (1) PLA is resistant to AD (Cazaudehore et al., 2022) and cannot be fully

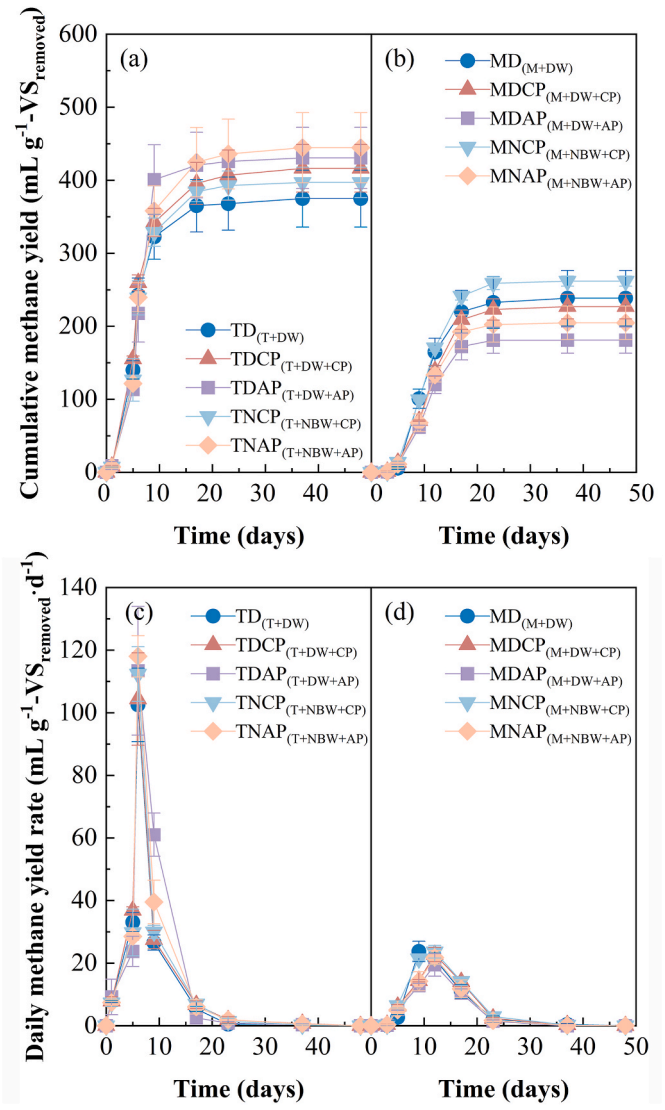


Fig. 3. Changes in CMY (a and b) and daily methane yield rate (c and d) during the digestion process.

hydrolysed under mesophilic conditions (Nie et al., 2024), which physically prevents microbes from entering a suitable environment to grow and metabolise (Lee et al., 2022); (2) the addition of MPs can reduce the extracellular polymer content of the sludge, while the reduction in extracellular polymers weakens the extracellular electron transfer process, reducing the methane yield (Hou et al., 2020; Tang et al., 2020; Wang et al., 2022b). The AP-added digester had less CMY than the CP-added digester, because aluminium as an outer coating, does not easily decompose into small particles and detach from the PLA plastic at 37 °C (Mwema et al., 2018). The aluminium film envelops the PLA plastic, preventing it from fully contacting the substrate and making it difficult for microorganisms to adhere to the PLA plastic for decomposition. Therefore, the aluminised film has a significant inhibitory effect on the hydrolysis of PLA under mesophilic conditions. In addition, the CMY was enhanced after the addition of NBW. It is speculated that the trace amount of oxygen generated by air-NBW promotes the hydrolysis of substrates in the acidification stage (Zhang et al., 2024), thus counteracting some of the resistance of PLA to AD under mesophilic conditions. The CMY in the MNCP group was the greatest as it was even more extensive than that in the control group. The rate of increase in CMY in the hydrolysis stage, from Day 5 to Day 9, of MNCP (Fig. 3d) was greater than that of all the other experimental groups. The MNCP was

Table 2
Parameters of the modified Gompertz model.

	Ultimate methane yield, UMY (mL g^{-1}-$VS_{removed}$)	Maximum methane yield rate, Rm (mL g^{-1}-$VS_{removed}$)	Lag phase, λ(d)	R^2
TD	366.1 ± 36.7^{b}	83.5 ± 7.7^{b}	3.2 ± 0.0^{d}	0.996 ± 0.001
TDCP	404.8 ± 31.8^{a,b}	71.8 ± 5.2^{b}	2.6 ± 0.2^{e}	0.994 ± 0.001
TDAP	426.0 ± 43.2^{a}	120.1 ± 21.0^{a}	4.2 ± 0.1^{c}	0.999 ± 0.000
TNCP	388.8 ± 18.6^{a,b}	77.9 ± 7.4^{b}	3.2 ± 0.4^{d}	0.994 ± 0.001
TNAP	433.1 ± 48.0^{a}	83.1 ± 12.8^{b}	3.3 ± 0.2^{d}	0.995 ± 0.002
MD	236.4 ± 36.6^{c,d}	27.6 ± 2.5^{c}	5.5 ± 0.1^{b}	0.999 ± 0.001
MDCP	229.2 ± 16.3^{c,d,e}	24.0 ± 2.0^{c}	6.0 ± 0.2^{a}	0.999 ± 0.000
MDAP	183.6 ± 17.9^{e}	21.0 ± 3.6^{c}	5.9 ± 0.5^{a}	0.999 ± 0.000
MNCP	263.8 ± 7.9^{c}	27.3 ± 0.4^{c}	5.4 ± 0.0^{b}	0.999 ± 0.000
MNAP	206.5 ± 24.8^{d,e}	23.6 ± 2.6^{c}	6.1 ± 0.4^{a}	0.998 ± 0.001

*Angular symbols "a", "b", "c", "d", and "e" indicate the difference between different working conditions. From "a" to "e" means that the value is getting smaller and smaller. The same letter indicates that the difference is not significant, while different letters suggest that the difference is significant.

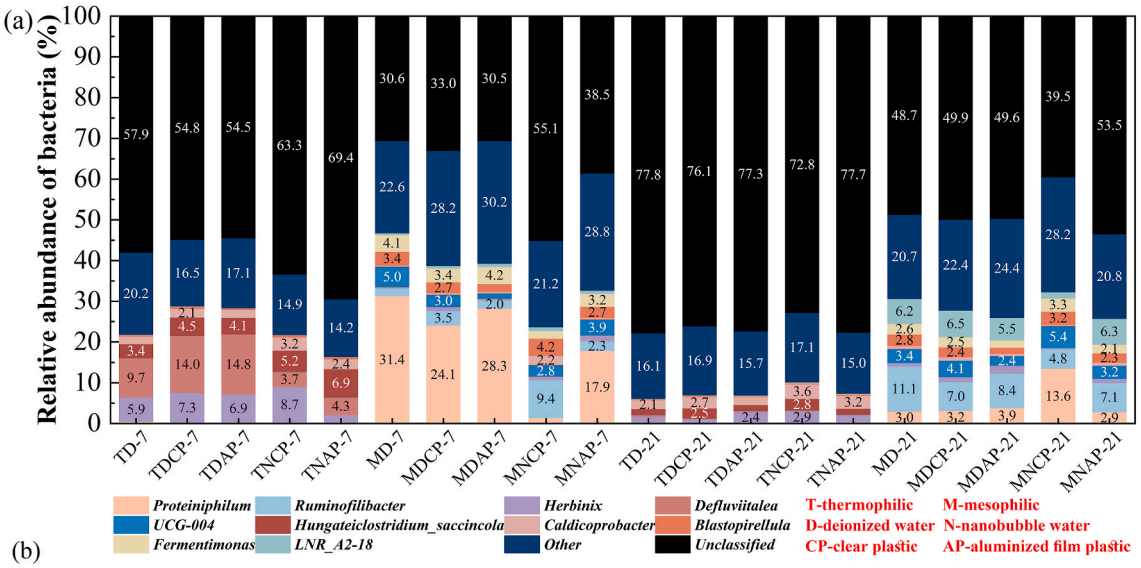
the only mesophilic experimental group with a λ value smaller than that of the control group, which shows that the effect of NBW on AD was slightly greater than that of MPs.

3.2. Characteristics of the microbial community

3.2.1. Dominant bacterial community

The relative abundance (RA) of bacteria is affected by temperature, MPs and NBW (Fig. 4). The RA of some bacteria, such as *Proteiniphilum* and *UCG-004*, was reduced under the influence of MPs on Day 7. The addition of MPs significantly reduces the extracellular polymer content of anaerobic granular sludge, and increased levels of MPs lead to decreases in the nucleic acid, polysaccharide, and protein levels (Wang et al., 2022b). *Proteiniphilum* is responsible for polysaccharide hydrolysis, especially the hydrolysis of cellulose and xylan (Wu et al., 2021b). *Proteiniphilum* has many types of sugar-degrading enzymes, including glycoside hydrolases, carbohydrate esterases, and substrate-specific carbohydrate-binding motifs (Basak et al., 2022) and shows a significant positive correlation with VFA levels ($P < 0.01$) (**see the Supplementary material section**). Therefore, the bacterial RA responsible for polysaccharide hydrolysis was reduced by the addition of MPs.

RA was elevated in some bacteria under the influence of MPs, such as *Ruminofilibacter*, *Herbinix*, *Defluviitalea*, *Hungateiclostridium saccincola*, *Caldicoprobacter*, and *LNR A2-18*. Except for *Defluviitalea* and *LNR A2-18*, the remaining four bacterial genus were more enriched in the CP-added group. *Ruminofilibacter* has hemicellulose activity to degrade hemicellulose, and its activity increases after cellulose hydrolysis to release hemicellulose (Yang et al., 2023). Therefore, hemicellulose is more likely to appear in the stage after hydrolytic acidification, resulting in increased RA in the later stage. The average RA on Day 21 was 50.5% greater than that on Day 7 and was negatively correlated with the SCOD level ($P < 0.05$). *Herbinix* is a thermophilic bacterium that can degrade lignocellulose and hemicellulose, produce VFA (Beri et al., 2020) and can promote the hydrolysis of cellulose and the acid production of organic matter (Wang et al., 2021b). Therefore, the RA in the late stage was 57.1% lower than that in the early stage, which was positively correlated with the SCOD level ($P < 0.05$). In the CM and CCW anaerobic



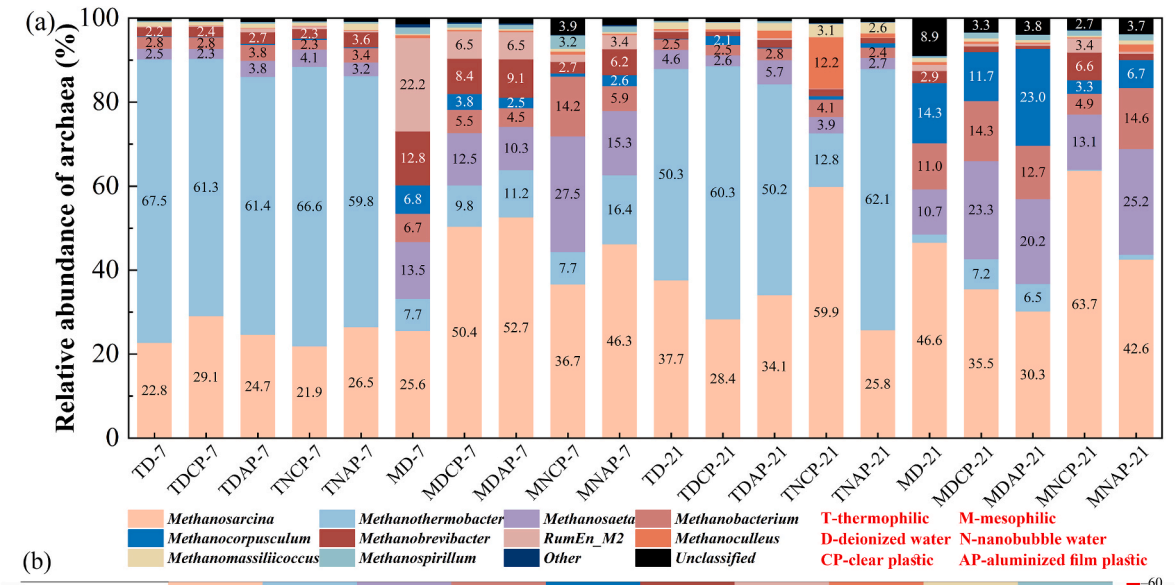
(b)

D-7	Test	Control	Protei-niphilum	Rumino-filibacter	Herbinix	Defluvii-talea	UCG-004	Hungatei-clostridium_ saccincola	Caldico-probacter	Blasto-pirellula	Fermen-timonas	LNR_A2-18
Temp	TD	MD	-30.9636	-1.7014	5.6239	9.5837	-4.9751	3.3631	1.6185	-3.0799	-4.0250	-0.4451
	TDCP	MDCP	-23.9032	-3.5219	6.4462	13.8553	-2.9538	4.3959	1.8395	-2.2261	-3.3603	-0.6141
	TDAP	MDAP	-28.0325	-1.9926	6.6002	14.6688	-1.3306	4.0478	1.5844	-1.5541	-4.1256	-0.7801
	TNCP	MNCP	-1.1804	-9.3401	8.0948	3.5538	-2.7079	5.0700	0.9737	-3.6683	-1.8041	-0.8937
	TNAP	MNAP	-17.6873	-2.2434	0.5707	4.1523	-3.7935	6.8166	2.0715	-2.2619	-3.1646	-0.7516
NBW	TNCP	TDCP	0.0177	0.0108	1.4415	-10.3066	0.0079	0.7301	1.0525	0.0705	0.0009	0.0141
	TNAP	TDAP	-0.0158	0.0065	-4.9975	-10.4822	0.0405	2.8519	0.5054	0.1156	-0.0067	-0.0074
	MNCP	MDCP	-22.7051	5.8290	-0.2071	-0.0051	-0.2380	0.0560	1.9183	1.5127	-1.5553	0.2937
	MNAP	MDAP	-10.3610	0.2573	1.0320	0.0343	2.5034	0.0831	0.0183	0.8234	-0.9677	-0.0359
CP	TDCP	TD	-0.2306	-0.0610	1.4482	4.3090	-0.0205	1.0587	0.3077	0.1626	-0.0418	-0.0126
	MDCP	MD	-7.2910	1.7595	0.6259	0.0374	-2.0418	0.0259	0.0867	-0.6912	-0.7065	0.1564
AP	TDAP	TD	-0.2016	-0.0708	1.0367	5.0798	-0.0393	0.6470	0.0928	0.0629	-0.0309	-0.0052
	MDAP	MD	-3.1327	0.2204	0.0604	-0.0053	-3.6838	-0.0377	0.1269	-1.4629	0.0697	0.3298

Fig. 4. Changes in bacterial genus during the digestion process(a), changes in RA of the bacterial community by different influence factors (b)
*The header color comparison legend in Fig. 4b represents different bacterial types. The red color in the table indicates that the variable increases the microbial RA, and the blue color indicates that it decreases it.

co-digestion system, the main sources of cellulose are straw and the cell wall tissues of CCW, which are not significantly affected by the negative impact of PLA-MPs. In contrast, MPs provide a larger specific surface area, offering more space for attachment (Liang et al., 2023), thus the bacterial RA involved in cellulose degradation increased. *Defluviitalea* can be selectively enriched under thermophilic conditions and plays roles in the oxidation of homotropic acetic acid, degradation of homotropic alcohols and acetic acid, degradation of complexes, and degradation of proteinaceous substances (Jiang et al., 2020). The RA in *Defluviitalea* was 97.1% greater in the early stage than in the late stage, showing a positive trend ($P < 0.05$) with SCOD. Under thermophilic conditions, the RA was 4.0%–5.0% greater with the CP-added digester than in the control group, which shows that CP can be considered to be externally added macromolecular organic matter under thermophilic conditions. AP-added digesters have a relatively high RA under thermophilic conditions, suggesting that the added conductive layer promotes the degradation of macromolecular organic matter, possibly because its conductivity facilitates interspecies electron transfer (Yu et al., 2022). Additionally, in mesophilic digesters with added CP, the RA of *Defluviitalea* increases, whereas in mesophilic digesters with added AP, the RA of *Defluviitalea* decreases. This suggests that under mesophilic conditions, CP undergoes slight degradation, whereas AP becomes a barrier to the degradation of macromolecular organic matter. Therefore, the inability of the applied aluminised film to break down at medium temperatures is hypothesised to hinder the interaction of microorganisms with PLA.

NBW influences the metabolic dynamics of anaerobic bacterial communities (Wang et al., 2024). *Proteiniphilum*, *Defluviitalea*, and *Fermentimonas* decrease in RA on Day 7 with the addition of NBW. It has been demonstrated that localised thermophilic, high-pressure, and hydroxyl radicals with vigorous oxidative activity during the explosion of nanobubbles produce a biocidal effect that leads to changes in community composition (Xiao et al., 2020). *Fermentimonas* is an acid-producing bacterium that enhances hydrolysis and acidification during AD (Basak et al., 2022). It thus has a higher activity at the earlier stage, which is manifested by the average RA on Day 7 being 26.1% greater than that on Day 21. In addition, *Proteiniphilum* and *Fermentimonas* are important synthesis partners of acetate-nutrient methanogens, and one of the primary metabolites they produce is acetate (Basak et al., 2022). Acetate-nutrient methanogens had larger RA at mesophilic (Fig. 5), so as their synthesis partners, they also had greater RA at mesophilic. *Defluviitalea* selectively enriches and degrades macromolecules under thermophilic conditions. These three bacteria function during the hydrolysis and acidification stages. The reduction in their RA indicates that the explosion of nanobubbles occurred during the hydrolysis and acidification phases, possibly causing the inactivation of these bacteria. Among them, *Proteiniphilum* and *Fermentimonas* also tended to have a decreasing RA after the addition of MPs. Therefore, it is hypothesised that the mobility of NBW led to a reduction in the aggregation of MPs and a more homogeneous distribution of MPs, which somewhat amplified the negative effects of MPs on them (Wang et al., 2023).



(b)

D-7	Test	Control	Methano-sarcina	Methano-thermobacter	Methano-saeta	Methano-bacterium	Methano-corpusculum	Methano-brevibacter	Rum-En_M2	Methano-culleus	Methano-massiliicoccus	Methano-spirillum
	TD	MD	-2.7969	59.7667	-10.9771	-3.8941	-6.6552	-10.6632	-22.0070	-0.6577	0.6662	-1.3066
	TDCP	MDCP	-21.3083	51.4579	-10.1539	-2.7186	-3.7138	-5.9770	-6.2975	-0.2440	0.2538	-0.7380
Temp	TDAP	MDAP	-27.9774	50.1861	-6.4752	-0.6995	-2.1986	-6.4074	-5.5985	-0.2018	0.6360	-0.6125
	TNCP	MNCP	-14.7725	58.8692	-23.4107	-11.9112	-0.5168	-0.4459	-1.0206	-0.5993	0.0093	-2.8677
	TNAP	MNAP	-19.7794	43.3683	-12.0606	-2.4755	-2.4370	-2.5846	-3.0485	-0.1301	1.1645	-0.9928
	TNCP	TDCP	-7.2109	5.3173	1.7672	-0.4655	0.1758	-0.1260	0.4720	0.0229	-0.1061	0.1094
NBW	TNAP	TDAP	1.7723	-1.5930	-0.5868	-0.3450	-0.0898	0.8641	-0.5810	0.0508	0.5791	-0.0116
	MNCP	MDCP	-13.7467	-2.0940	15.0240	8.7271	-3.0212	-5.6571	-4.8049	0.3782	0.1384	2.2391
	MNAP	MDAP	-6.4257	5.2248	4.9986	1.4310	0.1486	-2.9587	-3.1310	-0.0209	0.0506	0.3687
CP	TDCP	TD	6.3540	-6.2024	-0.2044	-0.0688	-0.0197	0.2439	0.0745	-0.0041	-0.1440	-0.0754
	MDCP	MD	24.8654	2.1064	-1.0276	-1.2443	-2.9611	-4.4423	-15.6350	-0.4178	0.2684	-0.6440
AP	TDAP	TD	1.9216	-6.0828	1.2880	0.9359	0.1695	0.5610	0.7295	0.0569	0.0458	0.1261
	MDAP	MD	27.1021	3.4978	-3.2139	-2.2587	-4.2871	-3.6948	-15.6790	-0.3990	0.0760	-0.5680

Fig. 5. Changes in archaea genus during the digestion process (a), changes in RA of the archaea community by different influence factors (b)
*The header color comparison legend in Fig. 5b represents different archaeon types. The red color in the table indicates that the variable increases the microbial RA, and the blue color indicates that it decreases it.

On Day 7, *Ruminofilibacter*, *Hungateiclostridium saccincola*, *Caldicoprobacter*, and *Blastopirellula* presented an increase in RA with the addition of NBW, which suggests that NBW acts as a facilitator in the hydrolysis stage and that its increased mobility and generation of reactive oxygen species promote the hydrolysis of lignocellulose, release hemicellulose, and increase the activity of these microorganisms with hemicellulolytic activity (Yang et al., 2020; Zhang et al., 2024). The trace level of oxygen in the environment produced by NBW facilitates direct interspecies electron transfer in most parthenogenetic bacteria, and has a positive impact on the hydrolytic acidification process during the entire AD process, providing more raw materials for the subsequent methanogenic phase (Wang et al., 2024).

3.2.2. Dominant archaeal community

Temperature, MPs and NBW also effected the RA of archaea (Fig. 5). The hydrogenotrophic methanogens presented the greatest RA under thermophilic conditions (34.2%–73.9%), followed by mixed-type methanogens (22.2%–60.7%), and the acetolactic methanogens presented the lowest RA (2.3%–5.7%). The greatest RA was observed in the mixed methanogens (32.3%–67.0%), followed by hydrogenotrophic methanogens (17.0–52.0%), with the acetolactic type having the lowest abundance (10.3%–27.5%).

The addition of MPs resulted in a decrease in the RA of some of the archaea. *Methanosaeta* and *Methanobacterium* exhibited decreased RA after the addition of MPs in most of the digesters, and *Methanothermobacter* also had a lower RA after the addition of MPs than did the control under thermophilic conditions on Day 7. *Methanothermobacter* is a hydrogenotrophic thermophilic methanogen (Pan et al., 2023). It is hypothesised that the degradation of PLA-MPs under thermophilic conditions enhances the activity of acetolactic methanogens and reduces the gene expression of hydrogenotrophic methanogens. The RA of *Methanothermobacter* is related to lactic acid conversion under thermophilic conditions and an increase in CMY from PLA (Clagnan et al., 2023). The highest CMY under thermophilic conditions was found in the TNAP, which had the greatest RA of *Methanothermobacter* throughout the AD. It can be hypothesised that MPs under thermophilic conditions may have participated in the process of AD as part of the substrate and that lactic acid monomers produced by the hydrolysis of PLA can be converted to biogenic methane (Clagnan et al., 2023).

Sometimes the addition of MPs also increases the RA in archaea. The RA of *Methanosaeta* and *Methanobacterium* increased in the AP-added digesters under thermophilic conditions. This may be because the aluminised film of AP fell off under thermophilic conditions, facilitating interspecies electron transfer of microorganisms due to its electrical conductivity (Yu et al., 2022). *Methanosaeta* and *Methanobacterium* produce methane through direct interspecific electron transfer (Pan et al., 2023; Zheng et al., 2021). *Methanomassiliicoccus*, as a methylotrophic methanogens, has been reported to be the most abundant AD-active archaea in the inhibited state (Cerrillo et al., 2018), whereas the RA of *Methanomassiliicoccus* was greater in the group of MPs-added digesters and was greater in the overall reaction at the later stage; thus,

MPs inhibited overall AD (Cazaudehore et al., 2022). The RA of *Methanosarcina* increase after addition of MPs on Day 7 (Fig. 5b), and the increase was more pronounced under mesophilic conditions (24.9%–27.1%). *Methanosarcina* is a hydrogenotrophic methanogen under high ammonia conditions, whereas it is an acetolactic methanogen under low ammonia conditions (Yi et al., 2023). Under thermophilic conditions, the TAN level exceeded 1500 mg L^{-1} on Day 7 (Fig. 1c), which reflects a high ammonia concentration. *Methanosarcina* is a hydrogenotrophic methanogen, whereas under mesophilic conditions, TAN was still at a lower level (Fig. 1d). *Methanosarcina* is an acetolactic methanogen. Under mesophilic conditions, the hydrophobic surface of MPs can aid in AD by adsorbing and retaining lipids and other hydrophobic compounds, thereby enhancing microbial metabolism (Lee et al., 2022).

The addition of NBW to the digesters also affected the RA of the archaea. On Day 7, under thermophilic conditions, the effect of NBW is not significant, possibly because the promotion effect of temperature on the entire AD process is already very pronounced; thus, the effect demonstrated by NBW is relatively small. Under mesophilic conditions, NBW causes a decrease in the RA of *Methanosarcina*, *Methanobrevibacter*, and *RumEn_M2*. It is possible that the localised high temperatures and pressures generated by the crushing of NBW affect their survival and change their community structure (Xiao et al., 2020).

The addition of NBW increased the RA of *Methanosaeta*, *Methanobacterium*, *Methanomassiliicoccus*, and *Methanospirillum* (Fig. 5b). Some studies report that as the TS decreases, the abundance of *Methanosarcina* decreases, whereas the abundance of *Methanosaeta* increases (Yan et al., 2022). *Methanosaeta*, an acetolactic methanogen, can survive well under high acetic acid conditions (Zhao et al., 2018), with greater accumulation of acetic acid in the digester, which is more suitable for growth. It was concluded that the addition of NBW facilitated the hydrolysis stage (Zhang et al., 2024), decomposed more macromolecules of organic matter, accelerated the decline in the TS content, and resulted in the accumulation of more acetic acid, decreasing that of *Methanosarcina* and increasing that of *Methanosaeta*. *Methanosaeta* was significantly negatively correlated with pH, TAN, FAN, and CMY ($P < 0.01$) (**see the Supplementary material section**). *Methanobacterium*, as a hydrogenotrophic methanogen, can perform direct interspecies electron transfer and is most abundant in electrochemically active aggregates (Zheng et al., 2021). The increase in the RA after the addition of NBW under mesophilic conditions enhances the methane yield under harsh environments. Overall, the addition of NBW decreased the RA of hydrogenotrophic methanogens by an average of 41.3%, increased the RA of mixed methanogens by an average of 21.7%, and increased the RA of acetate methanogens by an average of 16.9%.

4. Conclusions

PLA can be degraded in thermophilic anaerobic co-digestion reactions. PLA can also be used as an additional carbon source to release carbon-containing organic matter after degradation into lactic acid monomers and can increase the CMY of the AD system. The aluminised film on the surface of the PLA can be used as an electrically conductive material during AD, which promotes the CMY process. NBW did not have a significant effect on the degradation of PLA. CP and AP increased CMY by 10.6% and 16.5%, respectively, in the anaerobic co-digestion digester and by 6.2% and 18.3%, respectively, in the NBW-promoted anaerobic co-digestion digester. PLA did not play a promotional role in the mesophilic anaerobic co-digestion reaction but instead decreased CMY to different degrees. However, NBW was effective, increasing the CMY in the digester with CP-added by 11.6%. After the addition of PLA, lactate conversion and CMY in the thermophilic reaction were correlated with *Methanothermobacter*, so the reactor containing the most *Methanothermobacter* had the highest CMY. The addition of PLA under mesophilic conditions elevated *Methanomassiliicoccus* and on the basis of its elevated content in the inhibited state of the reactor, it was demonstrated that PLA inhibited AD under mesophilic conditions.

Funding

This work was supported by the National Natural Science Foundation of China [Grant No. 52360019].

CRediT authorship contribution statement

Peilin Guo: Writing – original draft, Visualization, Methodology, Formal analysis. **Tianfeng Wang:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Jie Wang:** Validation, Investigation, Formal analysis. **Jiazi Niu:** Validation, Investigation, Formal analysis. **Cheng Peng:** Investigation, Formal analysis. **Jiabei Shan:** Resources, Investigation. **Yu Zhang:** Supervision, Resources. **Haizhou Huang:** Supervision, Resources. **Jixiang Chen:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2024.143639.

Data availability

Data will be made available on request.

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